

Evidencias

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Tecnología en Análisis y Desarrollo de Software (ADSO)

Ficha: 3229209

Docker y Contenedores

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Video Evidencias:

Imagen 1

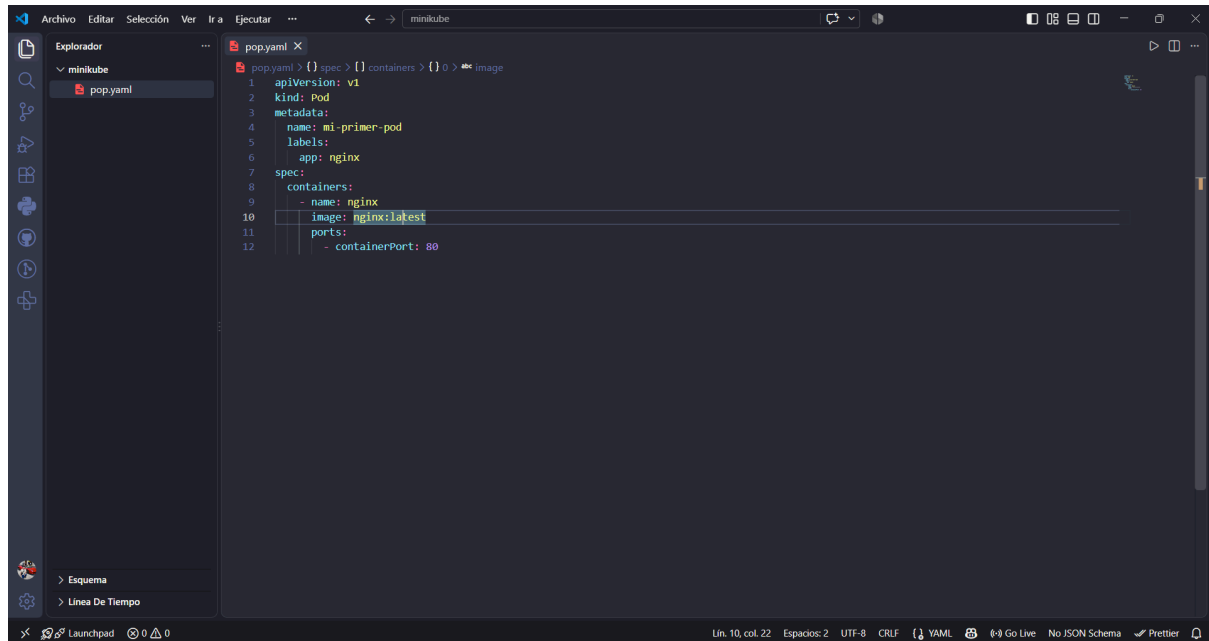


Imagen 2

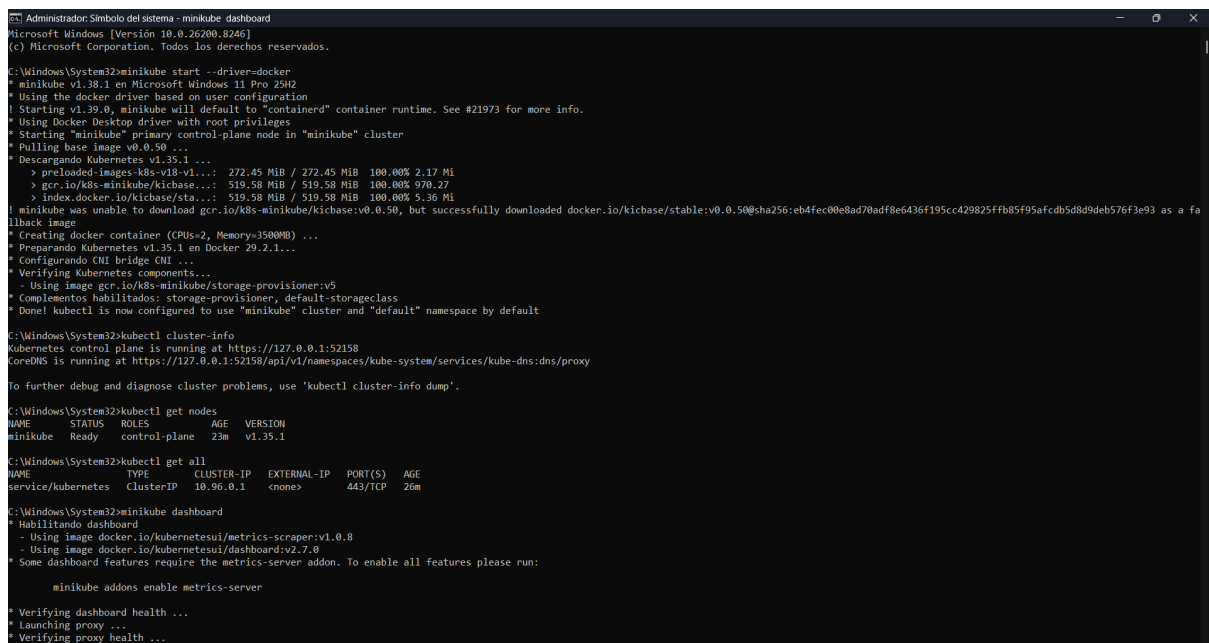


Imagen 3

```
Administrador: Símbolo del sistema - minikube dashboard
* Configurando CNI bridge CNI ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Complementos habilitados: storage-provisioner, default-storageclass
* Done! Kubectl is now configured to use "minikube" cluster and "default" namespace by default

C:\Windows\System32>kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:52158
coreDNS is running at https://127.0.0.1:52158/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use "kubectl cluster-info dump".

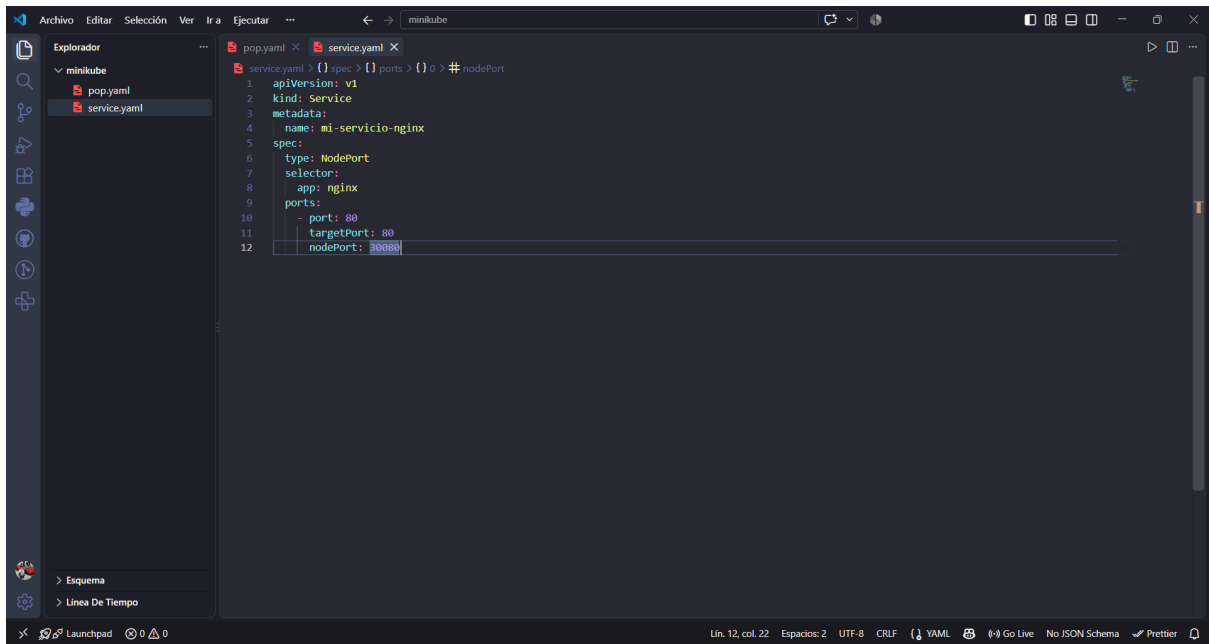
C:\Windows\System32>kubectl get nodes
NAME        STATUS    ROLES    AGE   VERSION
minikube    Ready     control-plane 23m   v1.35.1

C:\Windows\System32>kubectl get all
NAME                TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
service/kubernetes  ClusterIP     10.96.0.1    <none>        443/TCP    26m

C:\Windows\System32>minikube dashboard
* Habilitando dashboard
  - Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
  - Using image docker.io/kubernetesui/dashboard:v2.7.0
* Some dashboard features require the metrics-server addon. To enable all features please run:
    minikube addons enable metrics-server

* Verifying dashboard health ...
* Launching proxy ...
* Verifying proxy health ...
* Opening http://127.0.0.1:62925/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
```

Imagen 4



```
service.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: mi-servicio-nginx
5  spec:
6    type: NodePort
7    selector:
8      app: nginx
9    ports:
10     - port: 80
11       targetPort: 80
12     nodePort: 30080
```

Imagen 5

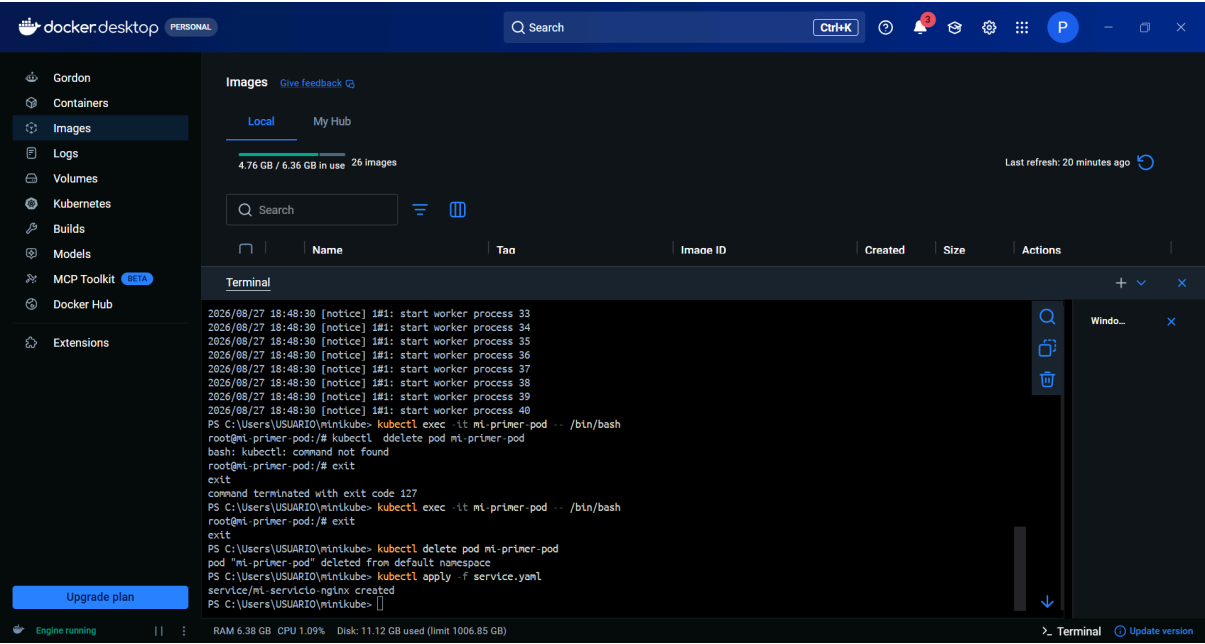


Imagen 6

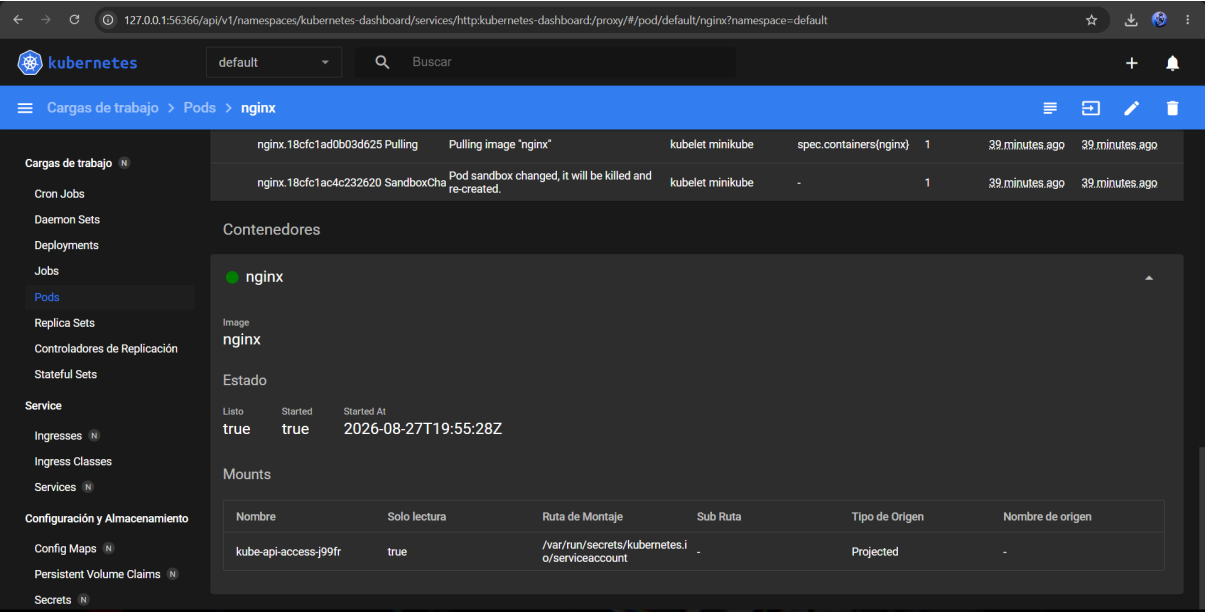


Imagen 7

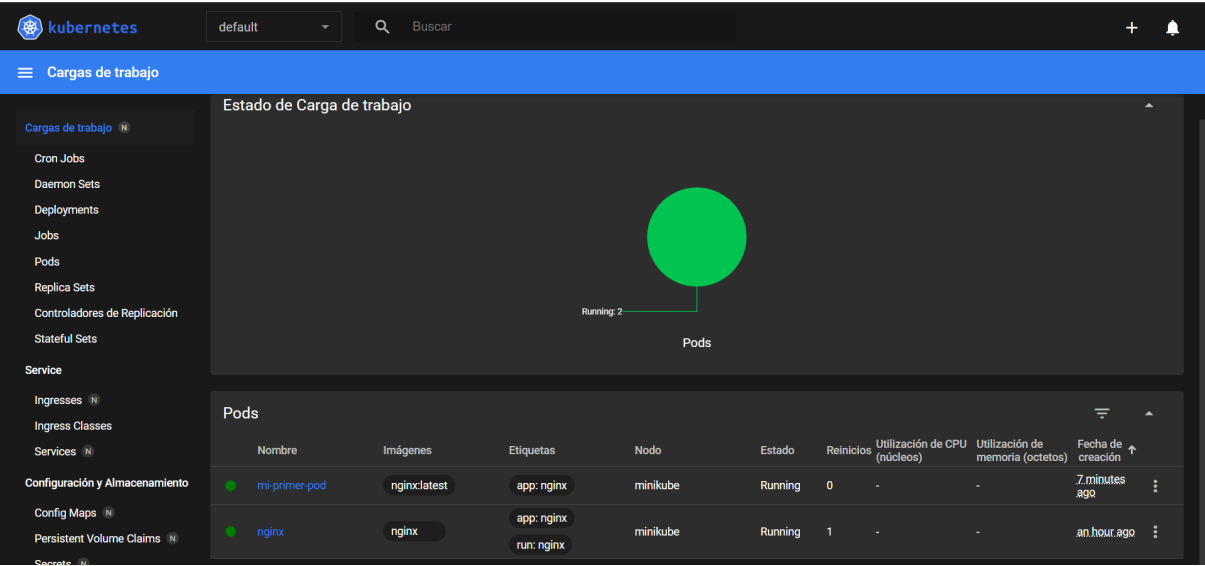


Imagen 8

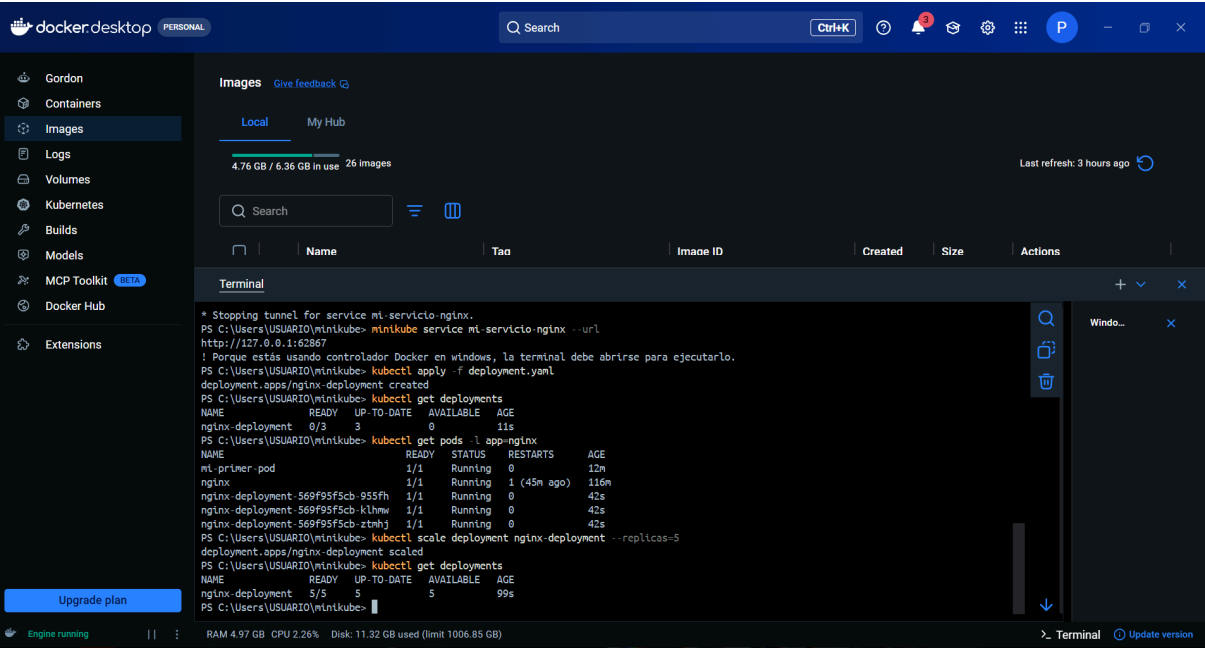


Imagen 9

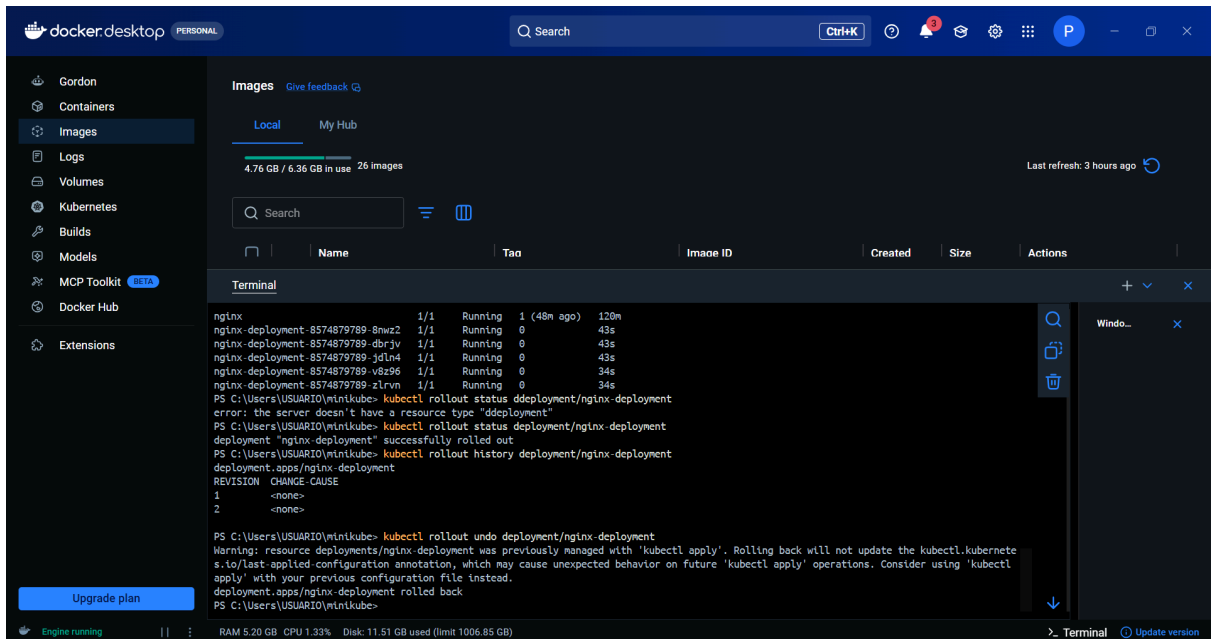
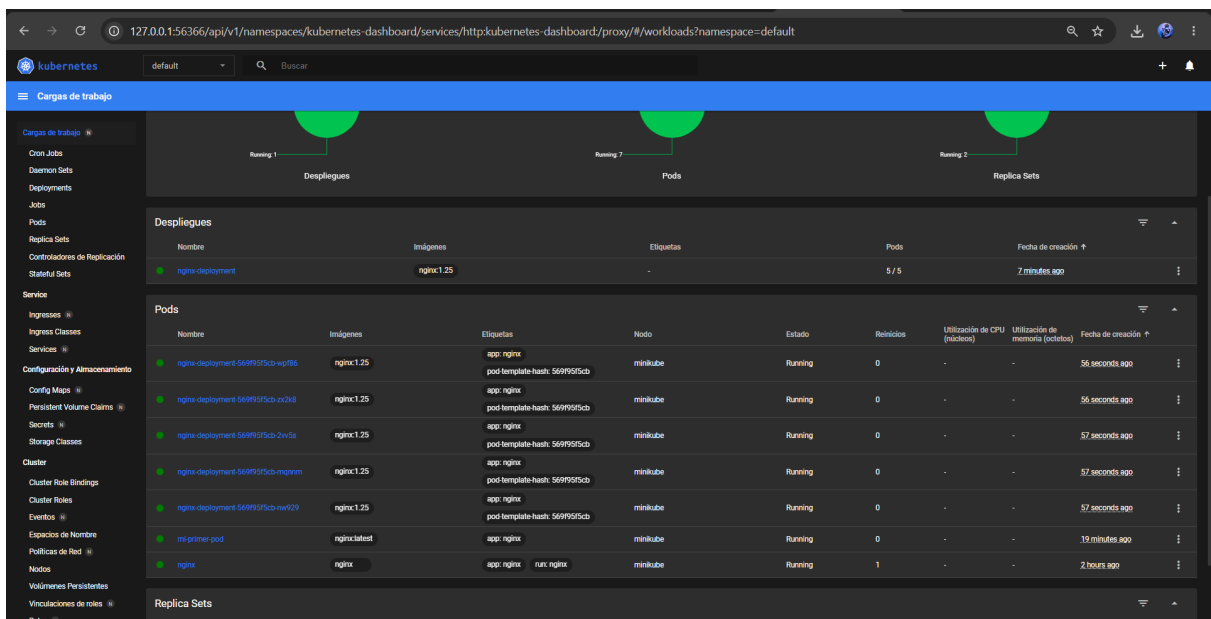
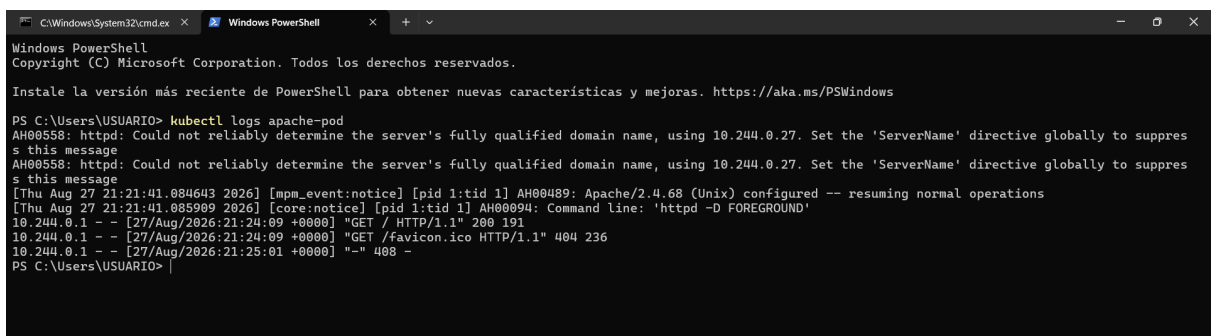


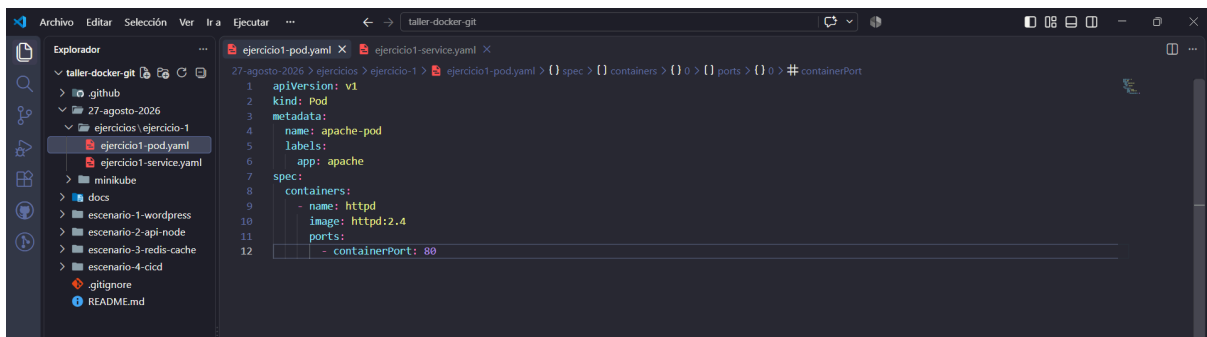
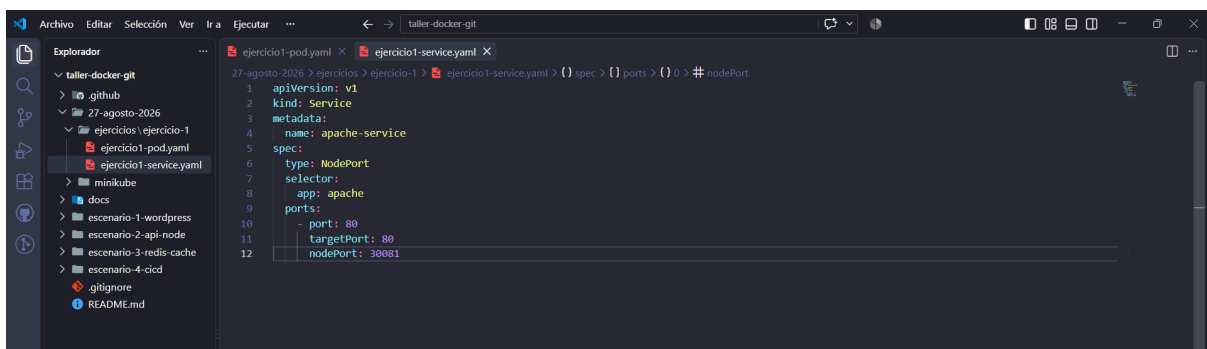
Imagen 10



Ejercicio 1 Evidencias:



```
C:\Windows\System32\cmd.exe x Windows PowerShell x + v
C:\Users\USUARIO\taller-docker-git>code .
C:\Users\USUARIO\taller-docker-git>cd 27-agosto-2026\ejercicios\ejercicio-1
C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-1>kubectl apply --dry-run=client -f ejercicio1-pod.yaml
pod/apache-pod created (dry run)
C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-1>kubectl apply -f ejercicio1-pod.yaml
pod/apache-pod created
C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-1>kubectl get pod apache-pod
NAME      READY   STATUS    RESTARTS   AGE
apache-pod 1/1     Running   0           21s
C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-1>kubectl get pod apache-pod
NAME      READY   STATUS    RESTARTS   AGE
apache-pod 1/1     Running   0           39s
C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-1>kubectl apply --dry-run=client -f ejercicio1-service.yaml
service/apache-service created (dry run)
C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-1>kubectl apply -f ejercicio1-service.yaml
service/apache-service created
C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-1>kubectl get service apache-service
NAME      TYPE      CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
apache-service NodePort   10.97.125.42  <none>        80:30081/TCP 15s
C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-1>minikube service apache-service --url
http://127.0.0.1:53294
! Porque estás usando controlador Docker en windows, la terminal debe abrirse para ejecutarlo.
```



Ejercicio 2 Evidencias:

```
C:\Windows\System32\cmd.exe x Windows PowerShell x + v
6b40cc5..c9138e4 docs\kubernetes-minikube -> docs\kubernetes-minikube

C:\Users\USUARIO\taller-docker-git>mkdir 27-agosto-2026\ejercicios\ejercicio-2

C:\Users\USUARIO\taller-docker-git>cd 27-agosto-2026\ejercicios\ejercicio-2

C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-2>kubectl apply --dry-run=client -f ejercicio2-deployment.yaml
deployment.apps/api-deployment created (dry run)

C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-2>kubectl apply -f ejercicio2-deployment.yaml
deployment.apps/api-deployment created

C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-2>kubectl get pods -l app=api
NAME READY STATUS RESTARTS AGE
api-deployment-57b845d4fb-cpsps 1/1 Running 0 25s
api-deployment-57b845d4fb-j8vvp 1/1 Running 0 25s
api-deployment-57b845d4fb-xlb6n 1/1 Running 0 25s

C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-2>kubectl apply --dry-run=client -f ejercicio2-service.yaml
service/api-service created (dry run)

C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-2>kubectl apply -f ejercicio2-service.yaml
service/api-service created

C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-2>kubectl get service api-service
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
api-service NodePort 10.103.237.206 <none> 80:30082/TCP 23s

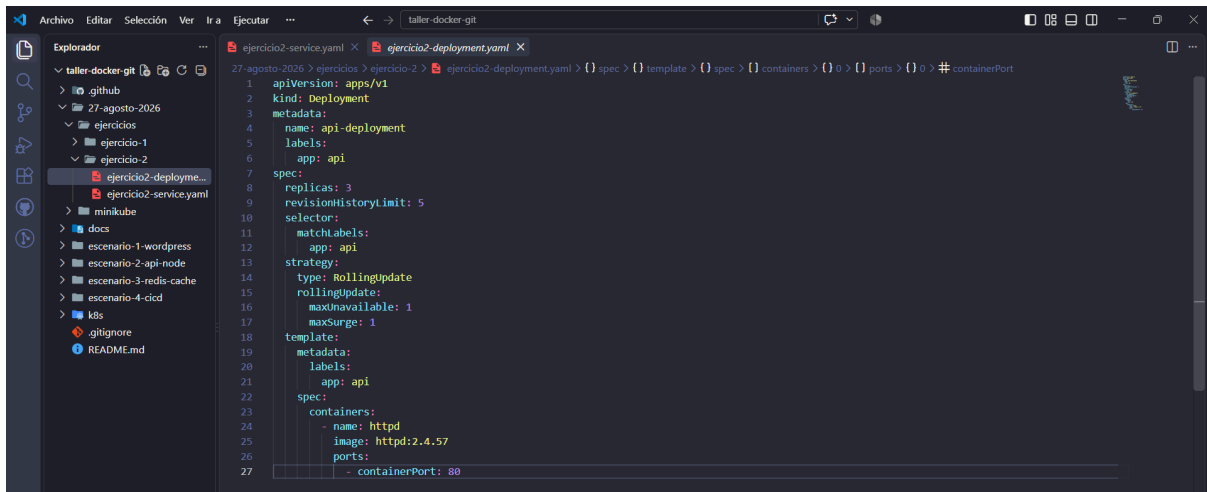
C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-2>minikube service api-service --url
http://127.0.0.1:55397
! Porque estás usando controlador Docker en windows, la terminal debe abrirse para ejecutarlo.
```

```
PS C:\Users\USUARIO> kubectl logs apache-pod
AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using 10.244.0.27. Set the 'ServerName' directive globally to suppress this message
AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using 10.244.0.27. Set the 'ServerName' directive globally to suppress this message
[Thu Aug 27 21:21:41.084643 2026] [mpm_event:notice] [pid 1:tid 1] AH00489: Apache/2.4.68 (Unix) configured -- resuming normal operations
[Thu Aug 27 21:21:41.085909 2026] [core:notice] [pid 1:tid 1] AH00904: Command line: 'httpd -D FOREGROUND'
10.244.0.1 - - [27/Aug/2026:21:24:09 +0000] "GET / HTTP/1.1" 200 191
10.244.0.1 - - [27/Aug/2026:21:24:09 +0000] "GET /favicon.ico HTTP/1.1" 404 236
10.244.0.1 - - [27/Aug/2026:21:25:01 +0000] "-" 408 -
PS C:\Users\USUARIO> kubectl get deployment api-deployment -o jsonpath="{.spec.template.spec.containers[0].image}"
httpd:2.4.57
PS C:\Users\USUARIO> kubectl set image deployment/api-deployment httpd=httpd:2.4.58
deployment.apps/api-deployment image updated
PS C:\Users\USUARIO> kubectl get deployment api-deployment -o jsonpath="{.spec.template.spec.containers[0].image}"
httpd:2.4.58
PS C:\Users\USUARIO> kubectl rollout status deployment/api-deployment
deployment "api-deployment" successfully rolled out
PS C:\Users\USUARIO> kubectl get deployment api-deployment -o jsonpath="{.spec.template.spec.containers[0].image}"
httpd:2.4.58
PS C:\Users\USUARIO> kubectl rollout history deployment/api-deployment
deployment.apps/api-deployment
REVISION CHANGE-CAUSE
1 <none>
2 <none>

PS C:\Users\USUARIO> kubectl rollout undo deployment/api-deployment
Warning: resource deployments/api-deployment was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-applied-configuration annotation, which may cause unexpected behavior on future 'kubectl apply' operations. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/api-deployment rolled back
PS C:\Users\USUARIO> kubectl rollout status deployment/api-deployment
deployment "api-deployment" successfully rolled out
PS C:\Users\USUARIO> kubectl get deployment api-deployment -o jsonpath="{.spec.template.spec.containers[0].image}"
httpd:2.4.57
PS C:\Users\USUARIO> kubectl get pods -l app=api
NAME READY STATUS RESTARTS AGE
api-deployment-57b845d4fb-49xnb 1/1 Running 0 59s
api-deployment-57b845d4fb-lqftm 1/1 Running 0 59s
api-deployment-57b845d4fb-sjlpw 1/1 Running 0 57s
```

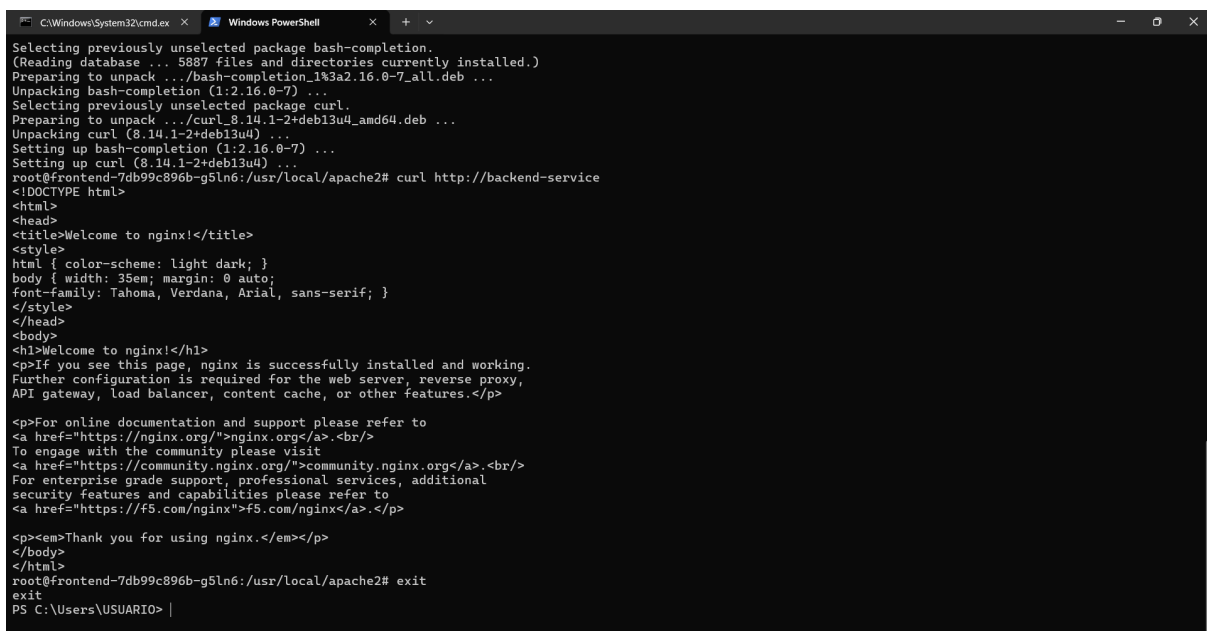
```
Archivo Editar Selección Ver Ir a Ejecutar ... taller-docker-git
Explorador
taller-docker-git
  .github
  27-agosto-2026
    ejercicios
      ejercicio-1
      ejercicio-2
        ejercicio2-deploye...
        ejercicio2-service.yaml
    minikube
  docs
  escenario-1-wordpress
  escenario-2-api-node
  escenario-3-redis-cache
  escenario-4-cicd
  k8s
  .gitignore
  README.md

ejercicio2-service.yaml x
27-agosto-2026 > ejercicios > ejercicio-2 > ejercicio2-service.yaml > {} spec > {} ports > {} 0 > # nodePort
1 apiVersion: v1
2 kind: Service
3 metadata:
4   name: api-service
5 spec:
6   type: NodePort
7   selector:
8     app: api
9   ports:
10     - port: 80
11       targetPort: 80
12       nodePort: 30082
```

It works!

Ejercicio 3 Evidencias:



```
C:\Windows\System32\cmd.exe x Windows PowerShell x + v
deployment.apps/backend created
service/backend-service created

C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-3>kubectl get pods -n mi-app -l app=backend
NAME READY STATUS RESTARTS AGE
backend-94c47b96f-hxkqn 1/1 Running 0 16s
backend-94c47b96f-n9csx 1/1 Running 0 16s

C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-3>kubectl apply --dry-run=client -f frontend-deployment.yaml
deployment.apps/frontend created (dry run)
service/frontend-service created (dry run)

C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-3>kubectl apply -f frontend-deployment.yaml
deployment.apps/frontend created
service/frontend-service created

C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-3>kubectl get all -n mi-app
NAME READY STATUS RESTARTS AGE
pod/backend-94c47b96f-hxkqn 1/1 Running 0 87s
pod/backend-94c47b96f-n9csx 1/1 Running 0 87s
pod/frontend-7db99c896b-g5ln6 1/1 Running 0 17s
pod/frontend-7db99c896b-k7sk4 1/1 Running 0 17s

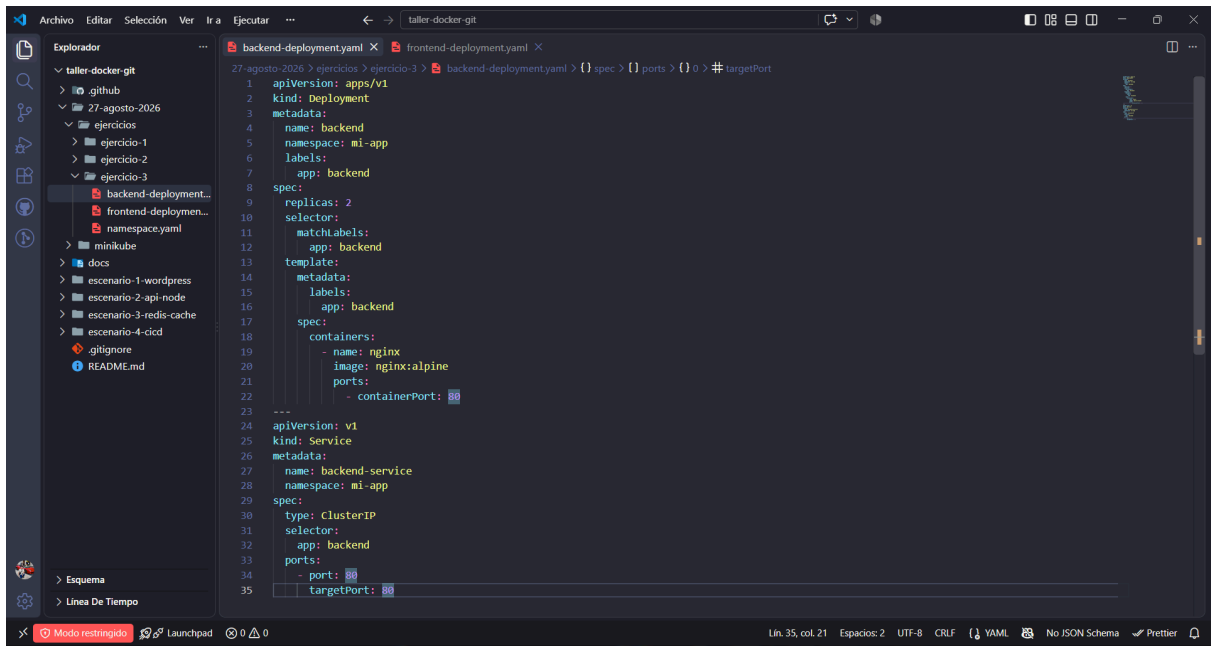
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
service/backend-service ClusterIP 10.103.37.281 <none> 80/TCP 87s
service/frontend-service NodePort 10.97.244.247 <none> 80:30083/TCP 17s

NAME READY UP-TO-DATE AVAILABLE AGE
deployment.apps/backend 2/2 2 2 87s
deployment.apps/frontend 2/2 2 2 17s

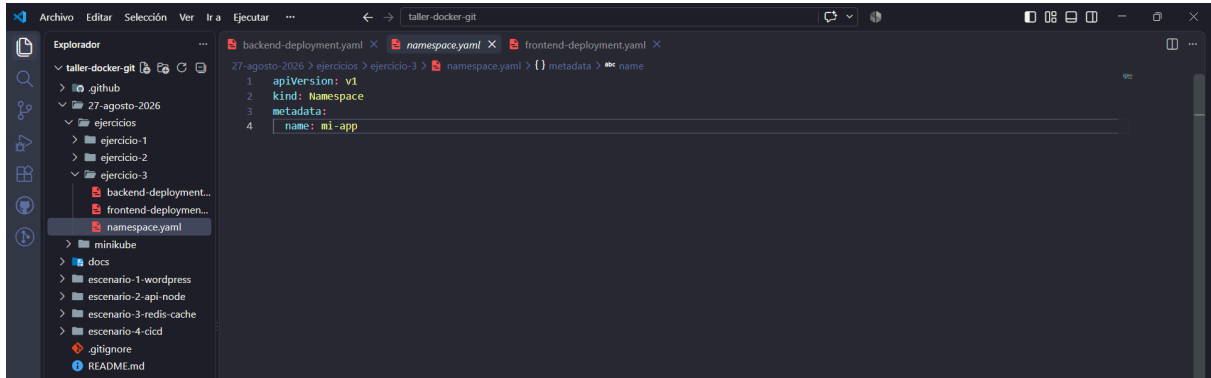
NAME DESIRED CURRENT READY AGE
replicaset.apps/backend-94c47b96f 2 2 2 87s
replicaset.apps/frontend-7db99c896b 2 2 2 17s

C:\Users\USUARIO\taller-docker-git\27-agosto-2026\ejercicios\ejercicio-3>minikube service frontend-service -n mi-app --url
http://127.0.0.1:51218
Porque estás usando controlador Docker en windows, la terminal debe abrirse para ejecutarlo.
```

```
27-agosto-2026 > ejercicios > ejercicio-3 > frontend-deployment.yaml > {} spec > {} ports > {} 0 > # nodePort
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: frontend
5    namespace: mi-app
6    labels:
7      app: frontend
8  spec:
9    replicas: 2
10   selector:
11     matchLabels:
12       app: frontend
13   template:
14     metadata:
15       labels:
16         app: frontend
17     spec:
18       containers:
19         - name: http
20           image: httpd:2.4
21           ports:
22             - containerPort: 80
23
24  apiVersion: v1
25  kind: Service
26  metadata:
27    name: frontend-service
28    namespace: mi-app
29  spec:
30    type: NodePort
31    selector:
32      app: frontend
33    ports:
34      - port: 80
35        targetPort: 80
36        nodePort: 30083
```



```
27-agosto-2026 > ejercicios > ejercicio-3 > backend-deployment.yaml > {} spec > {} ports > {} 0 > #targetPort
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: backend
5    namespace: mi-app
6    labels:
7      app: backend
8  spec:
9    replicas: 2
10   selector:
11     matchLabels:
12       app: backend
13   template:
14     metadata:
15       labels:
16         app: backend
17     spec:
18       containers:
19         - name: nginx
20           image: nginx:alpine
21           ports:
22             - containerPort: 80
23   ---
24   apiVersion: v1
25   kind: Service
26   metadata:
27     name: backend-service
28     namespace: mi-app
29   spec:
30     type: ClusterIP
31     selector:
32       app: backend
33     ports:
34       - port: 80
35       targetPort: 80
```



```
27-agosto-2026 > ejercicios > ejercicio-3 > namespace.yaml > {} metadata > {} name
1  apiVersion: v1
2  kind: Namespace
3  metadata:
4    name: mi-app
```